ENERGIA: USB-C, carga de bateria, reparto de carga y regulador 3.3 V

J1 — USB-C 16P (VBUS, CC1, CC2, USB_DM, USB_DP, SBU1, SBU2, SHIELD, GND)

U1 — TP4056 (CE, CHRG, STDBY, TEMP, PROG, BAT, VCC, EPAD, GND)

J2 — BAT+ / BAT−

Bateria LiPo soldada a pads.
Usar una con proteccion integrada

Q1 — AO3401A (VBUS, VBAT, VSYS)

U2 — AP2112K-3.3 (VSYS, PWR_EN, VIN, EN, VOUT, GND, 3V3_REG)

D3 — B5819W (VSYS, VBUS)

SW1 — ENCENDIDO (PWR_EN, VSYS, GND)

JP1 — 3V3 (3V3_REG, +3V3)

R1 5.1k CC1 · R2 5.1k CC2 · C1 10uF VBUS · R3 2k PROG · C2 10uF VBAT · R6 100k VBUS · C3 1uF VSYS · C4 10uF 3V3_REG · R15 100k PWR_EN · R11 1M VBAT · R12 1M BAT_SENSE · C11 100nF BAT_SENSE · R4 1k VBUS (LED_CHG_A) · R5 1k VBUS (LED_FULL_A) — todos a GND

D1 — CARGANDO (rojo) (CHRG_N, LED_CHG_A)

D2 — CARGA COMPLETA (verde) (STDBY_N, LED_FULL_A)

Puntos de prueba y banderas de alimentacion

TP1 VBUS · TP2 VBAT · TP3 VSYS · TP4 +3V3 · TP5 GND · TP6 EN · TP7 BTN_MENU · TP8 UART_TX · TP9 UART_RX

PWR_FLAG VBUS · PWR_FLAG VSYS · PWR_FLAG +3V3 · PWR_FLAG GND

PANTALLA (conector PROVISIONAL)

J3 — LCD 2.4in ST7789V (+3V3, GND, LCD_SCK, LCD_MOSI, LCD_CS, LCD_DC, LCD_RST, LCD_LEDA, LCD_LEDK)

Q2 — AO3400A (LCD_BL, LCD_LEDK, GND)

R13 10R (por definir) +3V3 → LCD_LEDA

R14 100k LCD_BL → GND

Pines y resistencia de luz de fondo:
se definen al tener el panel en la mano

microSD (SPI)

J4 — TF-01A (GND, SD_MISO, SD_CS, SD_MOSI, SD_SCK, DAT0(DO), DAT1(RSV), DAT2(RSV), CDDAT3(CS), CMD(DI), CLK(SCLK), VDD, CD, VSS)

R8 10k +3V3 → SD_CS · R9 10k +3V3 → SD_MISO · C8 100nF +3V3 · C12 10uF +3V3 — a GND

ESP32-S3-WROOM-1-N16R8

U3 — ESP32-S3-WROOM-1-N16R8 (3V3, GND, EN, BTN_MENU, BAT_SENSE, BTN_L, BTN_X, BTN_UP, BTN_DOWN, BTN_LEFT, BTN_RIGHT, BTN_A, LCD_DC, LCD_CS, LCD_MOSI, LCD_SCK, LCD_BL, LCD_RST, SD_MOSI, SD_SCK, UART_TX, UART_RX, SD_MISO, SD_CS, USB_DM, USB_DP, BTN_B, BTN_R, BTN_OPTION, I2S_BCLK, I2S_LRCLK, I2S_DIN, BTN_Y, BTN_START, BTN_SELECT)

SW2 — RESET (EN)

R7 10k +3V3 → EN · C5 1uF EN → GND · C6 10uF +3V3 → GND · C7 100nF +3V3 → GND

BOTONES (cada uno de su GPIO a GND, pull-up interno)

SW3 UP (BTN_UP) · SW4 DOWN (BTN_DOWN) · SW5 LEFT (BTN_LEFT) · SW6 RIGHT (BTN_RIGHT)

SW7 A (BTN_A) · SW8 B (BTN_B) · SW9 X (BTN_X) · SW10 Y (BTN_Y)

SW11 START (BTN_START) · SW12 SELECT (BTN_SELECT) · SW13 MENU (BTN_MENU) · SW14 OPTION (BTN_OPTION)

SW15 L (gatillo lateral) (BTN_L) · SW16 R (gatillo lateral) (BTN_R)

MENU (GPIO 0): presionado al encender entra en modo de grabacion.
Botones frontales: solo se usan dos pads en diagonal. L y R son de actuacion lateral.

AUDIO (I2S)

U4 — MAX98357A (I2S_DIN, I2S_BCLK, I2S_LRCLK, AMP_SD, VSYS, SPK_P, SPK_N)

J5 — BOCINA+ (SPK_P) · BOCINA− (SPK_N)

R10 1M AMP_SD → VSYS · C9 10uF VSYS → GND · C10 100nF VSYS → GND

GAIN sin conectar = 9 dB

Borrador generado. Sin probar en hardware.

Sheet: /
File: magsafe-console.kicad_sch

Title: MagSafe Console

Size: A2 | Date: | **Rev: 0.1**
KiCad E.D.A. 9.0.9 | Id: 1/1